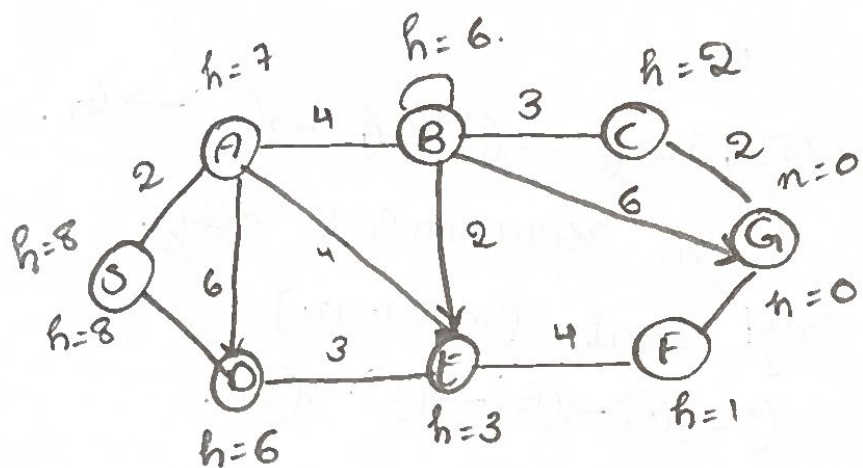


1. Smart Ambulance Navigation.

S. Mukarak

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AI [MLAO101].



by using $f(n) = g(n) + h(n)$.

starting node $\rightarrow S$, goal $\rightarrow G$.

1. S

$$g(S) = 0$$

$$h(S) = 8$$

$$\therefore f(S) = 0 + 8 = 8.$$

Expand $S \rightarrow A, D$.

A

$$g(A) = 2$$

$$h(A) = 7$$

$$f(A) = 2 + 7 = 9$$

$\therefore A \rightarrow$ [smallest].

Expand $A \rightarrow B, E, D$.

$$B \quad g(B) = 2 + 4 = 6. [\because g(A) = 2]$$

$$h(B) = 6.$$

$$f(B) = 6 + 6 = 12.$$

$$E \quad g(E) = 2 + 4 = 6$$

$$h(E) = 3$$

$$f(E) = 6 + 3 = 9$$

$$g(D) = 2 + 6 = 8$$

$$h(D) = 6.$$

$$f(D) = 12.$$

$\therefore E \rightarrow$ [smallest]

Path: $S \rightarrow A \rightarrow E$.

Expand $E \rightarrow B, F$.

$$g(F) = 6 + 4 = 10$$

$$h(F) = 1$$

$$f(F) = 10 + 1 = 11$$

$$B \quad g(B) = 6 + 2 = 8.$$

$$h(B) = 6.$$

$$f(B) = 8 + 6 = 14 \times$$

Expand $B \rightarrow C, G$.

$$C \quad g(C) = 6 + 3 = 9$$

$$h(C) = 2$$

$$f(C) = 9 + 2 = 11$$

$$g(G) = 6 + 6 = 12$$

$$h(G) = 0$$

$$f(G) = 12$$

$\therefore$ Path: $S \rightarrow A \rightarrow E \rightarrow F \rightarrow B \rightarrow C$

48

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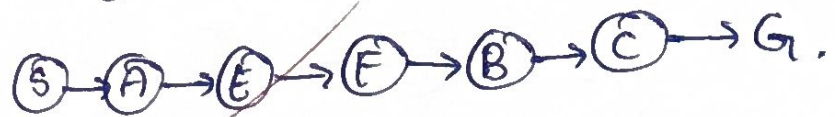
Expand C

$$g(G) = 9 + 2 = 11$$

$$h(G) = 0$$

$$\therefore f(G) = 11$$

$\therefore$ Selected path from S $\rightarrow$ G
is.



Total minimum cost = 11

Best route [minimum]



2. Autonomous warehouse Robot:

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

from given data,

we have

start: (5, 1) Goal (1, 7)

by using

$$h(n) = |x - x_g| + |y - y_g|$$

1. possible path.

(5, 1) $\rightarrow$ (5, 2) $\rightarrow$ (5, 3) $\rightarrow$ (5, 4) $\rightarrow$ (5, 5) $\rightarrow$ (5, 6) $\rightarrow$ (5, 7).

$\Rightarrow$ (5, 7) $\rightarrow$ (4, 7) Next (3, 7) is an obstacle.

2. Replanning the path,

(4, 7) $\rightarrow$ (4, 6) $\rightarrow$ (3, 6). [obstacle].

17

Again updating?

$$(3,6) \rightarrow (3,5) \rightarrow (3,4)$$

Then,

$$(3,4) \rightarrow (2,4) \rightarrow (1,4)$$

$$\text{finally } (1,4) \rightarrow (1,5) \rightarrow (1,6) \rightarrow (1,7)$$

$\therefore$ final path without any obstacles is

$$(1,4) \rightarrow (1,5) \rightarrow (1,6) \rightarrow G$$

3. University Exam Scheduling

a) csf formulation.

$$\text{Given, } \lambda = \{E_1, E_2, E_3, E_4, E_5, E_6\}$$

Each must be assigned a time slot.

Domains,

$$\text{initially, } D(E_1) = D(E_2) = \dots = D(E_6) = \{T_1, T_2, T_3, T_4\}$$

$\therefore$ Halt Capacities from data,

$$A=120, B=90, C=60$$

b) constraints

$$E_1 \neq E_2$$

$$E_1 \neq E_3$$

$$E_2 \neq E_4$$

$$E_3 \neq E_5$$

$$E_4 \neq E_6$$

$$E_5 \neq E_6$$

$\therefore$ Exam Remaining domain

$$E_1 \rightarrow \{T_2, T_3, T_4\}$$

$$E_2 \rightarrow \{T_1, T_2, T_3, T_4\}$$

$$E_3 \rightarrow \{T_2, T_3, T_4\}$$

$$E_4 \rightarrow \{T_2, T_3, T_4\}$$

$$E_5 \rightarrow \{T_1, T_2, T_3, T_4\}$$

$$E_6 \rightarrow \{T_1, T_2, T_3, T_4\}$$

∴ final Timetable

Time slot

Exam

Hall

students.

T_1

E_2

B

80

T_1

E_3

A

70

T_1

E_6

C

50

T_2

E_1

A

100

T_2

E_4

C

60

T_3

E_5

A

90

T_4

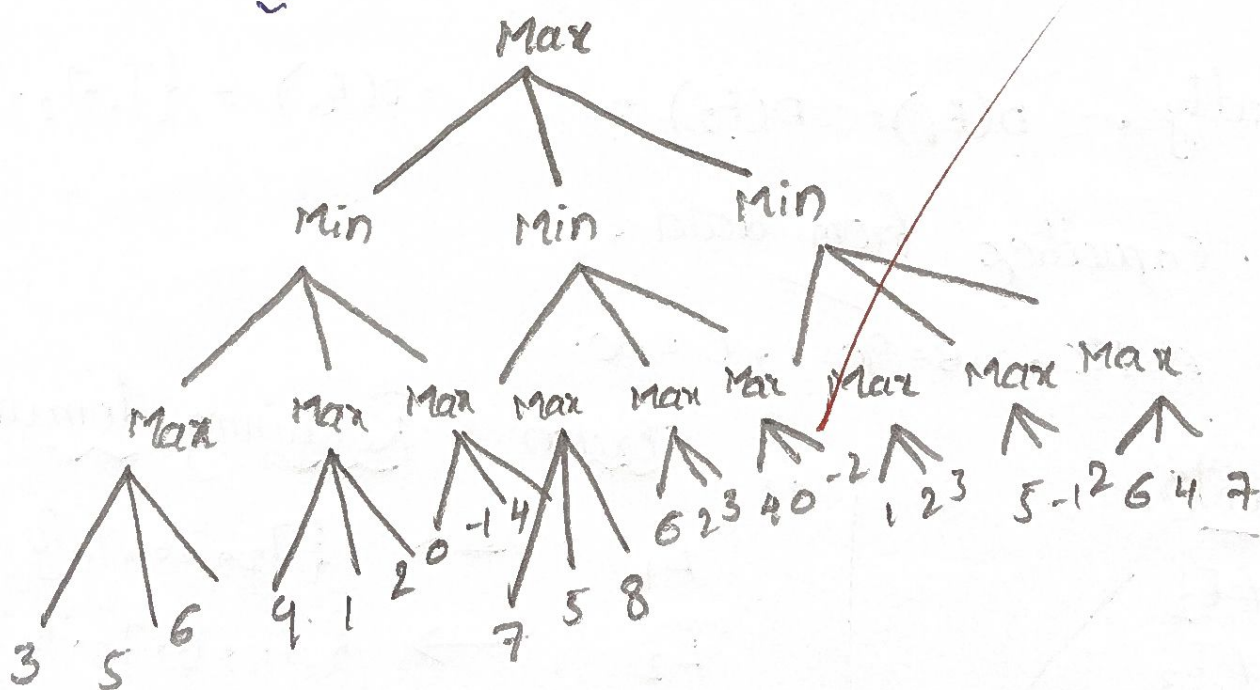
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∴ final Answer:

$E_1 = T_2, E_2 = T_1, E_3 = T_1, E_4 = T_2, E_5 = T_3, E_6 = T_1.$

4.

α - β Pruning?



1) MinMax calculation

Left

$$\text{Max}(3, 5, 6) = 6$$

$$\text{Max}(9, 1, 2) = 9$$

$$\text{Max}(0, -1, 4) = 4$$

$$\text{Min}(6, 9, 4) = 4$$

Middle

$$\text{Max}(7, 5, 8) = 8$$

$$\text{Max}(6, 2, 3) = 6$$

$$\text{Max}(4, 0, -2) = 4$$

$$\text{Min}(8, 6, 4) = 4$$

Right

$$\text{Max}(1, 2, 3) = 3$$

$$\text{Max}(5, -1, 2) = 5$$

$$\text{Max}(6, 4, 7) = 7$$

$$\text{Min}(3, 5, 7) = 3$$

$$\therefore \text{Max of } (4, 4, 3) = 4 //$$

optimum minMax value = 4 //

2) α - β pruning (left to right)

Pruning

Left

$$\text{Max}(3, 5, 6) = 6 \rightarrow \beta = 6.$$

$$\text{Max}(9, 1, 2) = 9, \alpha = 9$$

$$\alpha(9) \geq \beta(6) \rightarrow \text{prune}$$

$$\text{Max}(0, -1, 4) = 4.$$

$$\therefore \text{Min} = 4.$$

Middle

$$\text{values } (7, 5, 8) \rightarrow 8$$

$$(6, 2, 3) \rightarrow 6$$

$$(4, 0, -2) \rightarrow 4.$$

No pruning.

$$\therefore \text{Min value} = 4.$$

Right

$$\text{Max}(1, 2, 3) = 3 \rightarrow \beta = 3.$$

$$\text{Min} : \beta(3) \leq \alpha(4) \text{ prune.}$$

$$\text{Min} = 3 //$$

$$\therefore \text{Optimum minMax} = 4 //$$

(17)

⑤ Delivery route optimization.

Distance matrix (km)

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

neighbouring routes.

$$S_1: D \rightarrow A \rightarrow C \rightarrow B \rightarrow E \rightarrow D = 26 \text{ km}$$

$$S_2: D \rightarrow B \rightarrow A \rightarrow C \rightarrow E \rightarrow D = 30 \text{ km}$$

$$S_3: D \rightarrow A \rightarrow B \rightarrow E \rightarrow C \rightarrow D = 24 \text{ km}$$

$$S_4: D \rightarrow E \rightarrow C \rightarrow B \rightarrow A \rightarrow D = 27 \text{ km}$$

$$S_5: D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D = 23 \text{ km}$$

$$S_6: D \rightarrow C \rightarrow B \rightarrow A \rightarrow E \rightarrow D = 29 \text{ km.}$$

→ by comparing min is 23 km.

$$\therefore S_5: D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$$

neighbours of S_5 have cost 22 km.

$$\therefore \text{moves from } 23 \rightarrow 22 \Rightarrow \geq 22.$$

$$\therefore \text{final local search cost} = 22 \text{ km.}$$

cost of every neighbouring solution ≥ 22

final local optimum = 22 km